

A Study Evaluating AI Text Detection Models Across Datasets

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Declaration of AI: I declare that AI has been used as a support tool in this project. AI has been used to brainstorm project ideas, help refine the structure and grammar of the writing that I had produced. AI has been used to summarise or explain concepts from sources that I have struggled to understand. AI was also used to understand possible approaches to debugging specific errors in my application. Any suggested changes were independently verified, reviewed, tested, and understood by me before being finalised.

Abstract

With the rise of generative AI, AI-generated texts have become increasingly difficult to differentiate from human-written texts. Consequently, concerns about text origin and potential undermining of academic integrity have risen. To address these challenges, this project compares and evaluates the performance of classical and fine-tuned transformer models in detecting AI-generated text. All of the models were evaluated across two datasets and under a range of dataset settings, such as in-domain, cross-domain and combined domain. The two datasets were specifically chosen from the Hugging Face and Codabench platforms. In addition, the two classical models chosen for this project are Logistic Regression and LinearSVC, whilst the two fine-tuned transformer models are DistilBERT and DistilRoBERTa. Furthermore, to obtain an extended understanding of how fine-tuned transformer models behave, using the combined dataset, they underwent additional threshold-tuning and learning curve experiments. Overall, the results of this project indicated that whilst both transformer models achieved near-perfect performance under in-domain settings, their performance and generalisability dropped significantly across unseen cross-domain conditions. One notable observation in the HF→CB experiment was that both transformers exhibited severe one-class prediction bias. In contrast, although the classical models performed worse overall than the transformer models, they remained more stable, as they did not exhibit one-class prediction bias under cross-domain conditions. To build on this, by training all models on both datasets and introducing domain adaptation, their generalisability increased, and each model regained strong performance under combined-dataset testing conditions. In addition, the results for threshold tuning demonstrated minimal impact across both transformer models, indicating that the models were highly confident in their predictions. Moreover, results obtained from the learning curve experiment indicated that whilst increasing the dataset training size improved each transformer model's performance, the rate of improvement levelled off once the training sizes exceeded 50%. Overall, these results suggest that whilst current-day models can deliver strong performance in controlled environments, their performance and generalisability may be significantly reduced in real-world settings, as text is often noisy and subject to multiple evasion techniques. This suggests that a strong text detection system does not depend solely on a model's complexity, but also on the diversity of the datasets and evaluation strategies involved.

Introduction

In recent years, there has been a surge in the development of large language models, such as OpenAI's ChatGPT and Anthropic's Claude (Anthropic, 2023; OpenAI, 2022). These technologies have completely reshaped how written content is created, accessed, and consumed. As LLM technologies have become increasingly capable of producing coherent academic writing, distinguishing work created by humans and that produced by AI has become increasingly difficult (Fraser, Dawkins and Kiritchenko, 2025). Whilst LLMs can provide deeply enriching support, their generative abilities introduce significant risks, such as misinformation and academic misconduct (Fraser, Dawkins and Kiritchenko, 2025; Sadasivan et al., 2025). These challenges can target traditional plagiarism detection tools, which are designed only to identify similarities within existing sources; they cannot detect newly generated AI content, so work entirely produced by an LLM could appear authentic within those systems. This limitation highlights the increasing issue within academic institutions of verifying the authenticity of student submissions. As a result, concerns regarding student authorship have increased, posing a threat to academic integrity. In response to AI-generated content, AI detection tools have started emerging, such as GPTZero and ZeroGPT (ZeroGPT, 2023; GPTZero, 2023). However, many of these detection tools are limited in accuracy and generalisability, often producing false positives that flag written content as AI-generated (Fraser, Dawkins and Kiritchenko, 2025; Sadasivan et al., 2025). These limitations highlight significant weaknesses and issues in the current AI detection tools. As the field of AI text detection is new and constantly evolving (Kehkashan et al., 2025), it is evident that there is a gap in existing work regarding the reliability and generalisability of current models across different datasets and conditions (Dugan et al., 2024). Whilst some of these detection models can perform well in a controlled environment across a variety of conditions, they often struggle when confronted with unseen domains, generation, and more diverse real-world settings (Tufts, Zhao and Li, 2025).

This project aimed to evaluate the performance and generalisability of classical machine learning models and fine-tuned transformer models for AI text detection across multiple datasets and experimental conditions. Model performance was evaluated using standard classification metrics such as F1-scores, precision, recall and accuracy. This project aimed to make meaningful contributions to the field of AI by evaluating the performance and generalisability of classification models, determining their reliability beyond controlled environments, and assessing whether they are suitable enough for real-world deployment.

Contextual Review

Introduction to AI text Detection

AI-generated text can be defined as the text produced by an AI algorithm “in response to a user's prompt”, typically generated through LLMs (large language models) (Stryker and Scapicchio, 2024). On the other hand, AI text detection refers to the binary classification of determining whether a text was generated by an AI or a human (Fraser, Dawkins and Kiritchenko, 2025). In recent years, there has been increased interest in the field of AI, due to the widespread use of LLMs such as ChatGPT and Claude (Anthropic, 2023; OpenAI, 2022). Whilst these LLMs offer strong use cases for legitimate applications, they also raise concerns about data misinformation and academic misconduct (Fraser, Dawkins and Kiritchenko, 2025; Sadasivan et al., 2025).

Despite the aim of correctly classifying content authorship, achieving this is challenging because of the field's ever-changing nature. As LLMs become more fluent and capable of replicating human writing styles, the statistical differences between AI and human-written text decrease, making it difficult for detection models to be reliable. Furthermore, if the content generated by AI is either paraphrased or edited with human collaboration, the statistical features that enable the detection of AI can be removed altogether. As a result of these challenges, AI text detection has evolved into an increasingly active field of research. This literature review assesses multiple detection models, the effects of datasets on text detection systems, the evaluation methods used in this field, and the key limitations of existing research.

The Evolution of AI text Generation

The development of AI text generation has gone through four major phases, blurring the distinction between human and AI-generated content (Zhao et al., 2026). N-grams were one of the earliest classical NLP features which functioned by creating a “probabilistic model of word co-occurrence” that produced repetitive and fragmented text (Zhao et al., 2026). Following this, neural language models, such as recurrent neural networks (RNNs) and long short-term memory networks (LSTMs), were developed. These models functioned by producing texts with improved fluency through distributed word representations that captured a longer range of semantic relationships between words (Zhao et al., 2026). However, the most significant contribution stemmed from the introduction of transformer architectures. These architectures used a self-attention mechanism that enabled efficient modelling of relationships between words that were far apart in a text (Vaswani et al., 2017). Finally, by scaling these architectures into large language models such as ChatGPT or Claude AI, AI-generated text has become extremely coherent, to the point that even human reviewers struggle to reliably distinguish AI-generated news articles from authentic human-written articles (Brown et al., 2020).

The development of these models has a direct and important influence on the field of AI text detection; as the increasing quality of AI-generated text mirrors human capabilities, it also mirrors the statistical patterns and structure of human writing, resulting in reduced detection capability (Dugan et al., 2024; Fraser, Dawkins and Kiritchenko, 2025). Furthermore, decoding strategies and the ability to paraphrase or edit generated text mean that any possible remaining traces of AI-generated text can be obscured or removed altogether (Sadasivan et al., 2025).

Classical Machine Learning in AI text Detection

Classical machine learning functions by processing large amounts of data, identifying patterns within it, and, through this, predicting outcomes for unknown datasets (Bergmann, 2025b). In this regard, classical AI text detection models involve developing features by converting raw text into

numerical representations to distinguish between AI- and human-written text (Hua and Yao, 2024). Examples of such features include TF-IDF, N-grams, and Bag-of-words, which are often analysed alongside stylometric indicators such as word counts, constituency structure, and other general language characteristics (Schuster et al., 2020; Hua and Yao, 2024). These features are then provided to classifiers such as Logistic Regression, Support Vector Machine (SVM), and XGBoost to learn the key differences between machine- and human-written text (Hua and Yao, 2024).

Research studies evaluating several classical baseline models have suggested that SVM-based models typically outperform models such as Logistic Regression and XGBoost for detecting AI-generated content (Hua and Yao, 2024). This improved performance can be credited to the SVM-based models' ability to handle large numbers of text features, while avoiding overfitting to the dataset. In addition, studies often recognise classical models for their low computational costs, ease of interpretation, and simplicity (Hua and Yao, 2024). This suggests that classical models can serve as a baseline for understanding of the differences between AI-generated and human-written texts, enabling a broader comparative evaluation for more powerful models, such as transformers.

However, despite the advantages of classical machine learning models, they remain limited in many aspects, particularly when applied to different forms of text and contexts (Hua and Yao, 2024; Stiff and Johansson, 2022). For example, a classical model trained to detect features in long-form news articles will not be able to recognise those same statistical features in short-form text, such as social media posts (Stiff and Johansson, 2022). Additionally, shorter-form text can pose greater detection challenges to model performance in some settings (Maktabdar Oghaz et al., 2025). Finally, although these models can classify the content type of a given text, they cannot determine its contextual meaning or truthfulness. As generative language models have become more fluent and human-like over time, the limitations of these classical models have become more apparent, leading to the development of deep learning and transformer-based models.

Deep Learning Architectures in AI text Detection

Deep learning is inspired by the human brain; it uses artificial neural networks to learn from data automatically (Bergmann, 2025a). The introduction of deep learning architectures marked a broad transition in text classification from manually engineering features to automatically learnt textual representations (Conneau et al., 2017). While earlier approaches relied on stylometric indicators to detect machine-generated text, deep learning models enabled the automatic learning of textual representations from training data (Conneau et al., 2017; Schuster et al., 2020). This made them well-suited for text classification tasks that exhibited complex patterns that classical models could not detect. This transition led to the development of a range of deep learning detection models designed to identify different types of AI patterns. For example, Convolutional Neural Networks (CNNs) function by automatically identifying local patterns within a text frame and learning hierarchical representations, such as repeated phrasing or token-level patterns (Conneau et al., 2017). In contrast, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models process text sequentially, enabling them to model relationships between passages that are far apart (Conneau et al., 2017).

While deep learning models represent a substantial improvement over earlier baseline models in text classification (Howard and Ruder, 2018), studies have established that they still have important limitations. For example, deep learning sequential architectures such as RNNs and LSTMs struggle with parallel processing due to their sequential nature, making them less efficient than transformer models (Vaswani et al., 2017). Furthermore, studies indicate that neural network-based models can also be challenged by evasion techniques such as paraphrasing, and may not generalise well in out-of-distribution settings (Stiff and Johansson, 2022; Sadasivan et al., 2025). Overall, these deep

learning architectures act as a foundational bridge between classical models and the advanced transformer models used in this study.

Fine-Tuned Transformer Models in AI text Detection

A transformer model is a neural network architecture that understands contextual language by learning the relationships between words in a sentence (Winland, 2025). Transformer models such as Bidirectional Encoder Representations from Transformers (BERT) and the Robustly Optimised BERT Pretraining Approach (RoBERTa) have become prominent in NLP, as they offer strong performance across a wide variety of language tasks, partly due to their "self-attention mechanism" (Liu et al., 2019; Devlin et al., 2019). Unlike earlier architectures that process tokens sequentially, this mechanism enables transformers to capture complex, contextual relationships among all tokens in a sentence simultaneously (Devlin et al., 2019). BERT originally introduced this mechanism through masked language modelling (MLM) and next-sentence prediction (NSP). RoBERTa further refined the training procedure by removing NSP, introducing dynamic masking, and using a larger, more diverse pretraining data set (Liu et al., 2019).

Transformers are adapted for AI text detection by fine-tuning a pretrained transformer model on a dataset that contains both human and AI-generated text. During this process, an additional classification layer is added to the transformer model (Zellers et al., 2020). This layer uses a special classification token called [CLS]. The [CLS] token, placed at the start of an input sequence, acts as a summary for the entire input sequence. The resulting CLS token is then used to classify each data sample as either human-written or AI-generated (Priyadarshini, Chatterjee and Rawat, 2026). This approach is advantageous because the model uses transfer learning, which allows the general language knowledge gained from pretraining to be applied to AI detection tasks (Priyadarshini, Chatterjee and Rawat, 2026).

Studies generally report that fine-tuned transformers outperform both classical and deep learning models, achieving very high benchmark performance that sometimes exceeds 99% accuracy (Priyadarshini, Chatterjee and Rawat, 2026). The success of these transformers is often attributed to their ability to understand contextual relationships within texts. These exceptional performances should, however, be interpreted with caution, as research using TURINGBENCH has shown that while transformer-based detectors can perform well at detecting AI-generated text in a controlled domain, their F1-scores drop significantly when encountering text produced by sophisticated text generators that closely resemble human writing (Uchendu et al., 2021).

A further limitation of transformer-based models is their high vulnerability to overfitting when encountering specific dataset artefacts or patterns (Kuznetsov et al., 2024; Borile and Abrate, 2025). This occurs when transformer models achieve high accuracy within a dataset by picking up surface-level patterns, rather than learning generalisable features that can distinguish between AI and human text in other datasets (Kuznetsov et al., 2024). Building on this observation, research papers have suggested that transformer-based models can often lack this generalisability. For example, the transformer-based models trained on one domain type can struggle to generalise well to other domain types, with different topics and writing styles (Kuznetsov et al., 2024). This can result in significant performance degradation. For this research project, the generalisability of the transformer models was assessed not only in-domain but also across domains with unseen text styles.

The Role of Datasets within AI text Detection

Datasets are a fundamental component of AI text detection; they consist of both human-written and AI-generated texts spanning a variety of domains, such as reviews, news articles, and social media

content (Uchendu et al., 2021). These domains can also differ in sentence length, writing style, and topic (Borile and Abrate, 2025). For example, the TURINGBENCH dataset is composed of news-style articles that include both AI-generated text from various models and human-written content (Uchendu et al., 2021). In contrast, the HC3 dataset domain consists of various human-written and AI-generated question-and-answer-style content (Guo et al., 2023). Research has shown that the diversity of the dataset used for training and testing AI models significantly impacts their observed performance, especially when assessing their generalisability across different forms of text (Borile and Abrate, 2025).

Datasets can also pose limitations, as research shows that AI detection models can often exploit artefacts/surface-level features that work well with the dataset on which they are trained, rather than identifying useful features that can be utilised for generalisation across different datasets (Borile and Abrate, 2025). Another limitation presented across various datasets is the limited representation of short-form writing. Longer-form writing is usually preferred because it provides a stronger context and clearer structure for a model to learn from. However, this is a problem as short-form writing is generally more challenging to classify (Kehkashan et al., 2025). To further investigate these limitations, this research project evaluated both in-domain and cross-domain tests across a variety of classical and transformer models.

Evaluation Metrics within AI text Detection

Models are often evaluated using metrics such as precision, accuracy, recall, F1-score and confusion matrices. Confusion matrices consist of four different possible outcomes: True positives, True negatives, False positives, and False negatives (Saito and Rehmsmeier, 2015). Accuracy measures the total proportion of correct classifications; however, this metric can be limited, as in cases involving an imbalanced dataset with a greater split toward AI texts, or vice versa, accuracy may yield inflated results even when the data is misclassified (Saito and Rehmsmeier, 2015). As a result, metrics such as precision (the proportion of correctly predicted positives among all predicted positives), recall (the proportion of actual positives correctly identified), and F1-score are used to provide a more balanced evaluation of a model's performance (Saito and Rehmsmeier, 2015).

In addition, research suggests that popular metrics such as ROC-AUC (a measure of how well a model distinguishes between classes at different thresholds) can yield inflated results under imbalanced datasets (Saito and Rehmsmeier, 2015). In contrast, precision-recall curves can provide stronger evaluation readings, as they focus on how well the model handles the positive class and the impact of false positives (Saito and Rehmsmeier, 2015). The evaluation of these text classification models should extend beyond standard evaluation metrics by considering textual changes such as domain and generation style, as models that perform well in controlled environments may not generalise effectively in real-world settings (Dugan et al., 2024). Therefore, this project used a variety of evaluation metrics whilst also examining model behaviour across different dataset types.

Challenges and Research Gaps In AI text Detection

The field of AI text detection is inherently challenging because generative AI models are continually improving their ability to produce human-like, stylistic writing, thus making it increasingly difficult for detection tools to distinguish AI-generated text from human-written text (Kehkashan et al., 2025). These challenges are further emphasised, as detection tools can be vulnerable to a variety of evasion techniques, such as paraphrasing and human editing, undermining the reliability of these models' detections (Sadasivan et al., 2025).

In addition to these broader challenges, current AI text detection methods have their own limitations. For example, supervised learning models and fine-tuned architectures can perform well in controlled environments that they are trained on; however, they can fail to generalise across

different domains, indicating poor performance in real-world settings (Kehkashan et al., 2025; Tufts, Zhao and Li, 2025). Watermarking-based approaches can achieve high accuracy in detecting AI-generated text, but their effectiveness can be weakened by evasion techniques such as paraphrasing (Sadasivan et al., 2025). Zero-shot detection approaches can often generalise better than supervised-learning methods in some unseen domains; however, they are limited by their computationally-intensive nature (Mitchell et al., 2023). Therefore, these limitations can prevent them from operating effectively in real-world environments.

There is a clear gap within the current research space due to the lack of rigorous comparative testing across datasets, models, and real-world settings (Dugan et al., 2024). This is an issue, as it remains unclear which detection methods generalise best in real-world use (Tufts, Zhao and Li, 2025). In addition, it is worth noting that current research on AI text detection has not comprehensively examined model performance against text that has been rewritten, paraphrased or modified (Dugan et al., 2024). Furthermore, many of those studies have relied on overall detector performance measures rather than providing a descriptive analysis of a detector's false-positive rates (Dugan et al., 2024; Tufts, Zhao and Li, 2025).

Conclusion

The evolution of AI text detectors has gone from classical machine learning to deep learning models, and then to transformer-based architectures, with each development introducing a new dimension in detecting AI-generated text. Models like BERT and RoBERTa have been found to achieve strong benchmark performance on static text for AI-text datasets. However, as language models have continued to evolve, so have the methods for generating machine-generated text that can evade AI text detectors. No single methodology for detecting AI-generated text has yet emerged that is robust across all domains, text generators, and text lengths/styles. Furthermore, many of the evaluations of AI text detectors do not appear to be consistent with one another. Thus, the purpose of this project is to evaluate existing models for detecting AI-generated text and to determine the most reliable text-detection methodologies.

Methodology and Project Process

Methodology Overview

This research employs a comparative experimental methodology to evaluate the performance of baseline classical and fine-tuned transformer-based models for AI text classification. This comparative research was carried out to systematically investigate differences in the capabilities, generalisation, and behaviours of the models under diverse dataset conditions. This experiment employs two TF-IDF-based classical models: Logistic Regression and LinearSVC. These classical models served as a baseline for evaluating AI text classification more broadly. In addition, fine-tuned transformer models such as DistilBERT and DistilRoBERTa were utilised to capture deeper contextual relationships in text classification that baseline models cannot. These models were evaluated on two distinct datasets sourced from Codabench and Hugging Face (Lamsiyah et al., 2025; ahmadreza13, 2025). Each dataset consists of binary classification labels that represent human-written and AI-generated text. These datasets were used to assess model performance across varying conditions, such as in-domain, cross-domain, and combined-domain settings, to evaluate the generalisability of each model.

Research Objectives and Questions

The main research objective of this project is to evaluate the performance of classical and fine-tuned transformer models under varying test conditions across datasets. The models were tested on their classification ability through in-domain evaluation and their ability to generalise across datasets with different language styles and domains. Furthermore, an extended and comprehensive performance analysis of the models was conducted using a combined dataset. In addition, transformer models underwent model sensitivity tests, such as threshold tuning and learning curves, to understand how model behaviour is affected by decision thresholds and varying training dataset sizes.

To address these objectives, the following research questions have been established:

1. How do TF-IDF feature-based classification models compare in performance against fine-tuned transformer models in all related AI text classification experiments?
2. How well do all classical and transformer models generalise across different datasets when tested in cross-domain settings, such as training on Hugging Face and testing on Codabench?
3. To what extent does a varying dataset training size impact model performance across both fine-tuned transformer models, and why?
4. To what extent does the threshold tuning experiment affect the predictive behaviour across both fine-tuned transformer models, and why?
5. What do the error patterns that occur across the best and worst performing models indicate?

Requirements and Success Criteria

The functional requirements for this research established a structured approach to ensure research validity is maintained at all times. The key functional requirements are as follows:

1. The system must support the implementation and training of TF-IDF-based classical models such as Logistic Regression and Support Vector Machines.
2. The system must support the implementation and training of fine-tuned transformer models, such as DistilBERT and DistilRoBERTa.
3. The system must support the loading and handling of datasets, including Hugging Face, Codabench, and the combined dataset, for model training and testing.
4. The systems must successfully preprocess and clean datasets, ensuring there is no direct data leakage between the training and test sets.
5. The system must successfully execute each experiment for both classical and transformer-based models, including in-domain, cross-domain, combined-domain, threshold-tuning, and learning-curve experiments.

The success of this project depends on its ability to generate meaningful comparative evaluations of each model's generalisation and performance across diverse experimental conditions, including in-domain, cross-domain, and combined-domain settings. To maintain research validity, all of the experiments were conducted using consistent settings, such as the standardisation of dataset sizes; for example, single dataset evaluations consisted of 20,000 records and combined dataset experiments consisted of 40,000 records. Alongside this, uniform data preprocessing procedures and the use of consistent evaluation metrics (confusion matrices, accuracy, F1-score, precision, and recall) were applied across all models.

Tools and Technologies

This project utilised Python as the primary programming language, alongside a Jupyter Notebook IDE. The combination of these programming tools offers a wide variety of NLP libraries, as well as efficient code experimentation and debugging. Python libraries such as scikit-learn were used to implement classical models including Logistic Regression and LinearSVC, using TF-IDF feature extraction. TF-IDF has been chosen as the primary feature extraction technique, due to its simplicity, computational efficiency and performance capability for text classification. Furthermore, Hugging Face's transformer library was used alongside PyTorch, to implement, train and evaluate fine-tuned transformers such as DistilBERT and DistilRoBERTa. Finally, Python libraries such as NumPy and Pandas were used to support an effective dataset preprocessing and manipulation.

Project Planning and Management

The plan for this project has been divided into five core phases: Contextual background research, selection of datasets, selection of classical models, selection of fine-tuned transformer models, and workflow development and execution. The sequence of these phases have been intentionally selected, so that each following phase is supported by the knowledge built from previous stages.

The contextual background research was a crucial phase in the project plan, as it contributed to the development of a strong knowledge framework by identifying key concepts and challenges in the field of AI text classification. This process helped establish a clear set of criteria, which was used to select the most suitable classification models and datasets for this project. Instead of selecting tools first and then building the project around them, a contextual background review was conducted as it allowed project objectives to be established, so later decisions could be effectively aligned with the objectives.

In the project plan, dataset selection served as a critical phase, as the quality of a selected dataset can directly influence the project's ability to fulfil the established research objectives. Furthermore, the characteristics of a dataset can directly impact a model's performance and their ability to generalise across unseen datasets. The first dataset was successfully selected from the Hugging Face platform due to its compatibility with the project objective. Following this, another dataset was identified and selected from a website due to the advertised content type, record size and licensing availability, however upon close examination, the dataset had contained multiple mismatched discrepancies, such as the limited record quantity and class imbalance. This dataset could have been a major limitation to the project if chosen without inspection, as it could have potentially harmed future model performance, generalisability and comparative evaluations. To overcome this complication, a meeting with the supervisor was organised, resulting in a suitable dataset from Codabench being recommended. This resolution improved the reliability of the experimental pipeline, ensuring the research objectives were maintained.

Following the selection of the datasets, the classification models were carefully identified and chosen based on their ability to serve as baselines representing different levels of classification complexity for comparative analysis against more complex models. The contextual background research led to the selection of Logistic Regression and LinearSVC models. These classical models were chosen as a result of their simplicity and strong compatibility with TF-IDF text classification. Furthermore, for the complex models, two fine-tuned transformer models were selected: DistilBERT and DistilRoBERTa, which were chosen instead of their full-size base models as they provide similar performance at a lower computational cost. This decision positively impacted development efficiency as it enabled faster iterative experimentation with complex models.

The last phase consisted of developing an iterative model workflow pipeline to streamline the training and experimentation of the models at the implementation stage. This enabled efficient model training and evaluation to occur across all the planned experiments, with all evaluation metrics being systematically recorded for later analysis and comparison. As this project utilised an iterative workflow for all the experiments, every error and unexpected result was examined and utilised to re-develop the workflow pipeline and model configurations.

The progress of this project was monitored using a variety of tracking tools. Weekly progress logs were utilised to record progress, major project decisions and any challenges that were encountered. Two primary scheduling tools such as Trello and Gantt Charts were employed to track project milestones and manage task deadlines (see Appendix A and Appendix B). These progress tracking tools were extremely important as they helped manage deadline issues such as unexpected setbacks or recurring schedule conflicts due to university coursework or exams.

Implementation Strategy & Justification

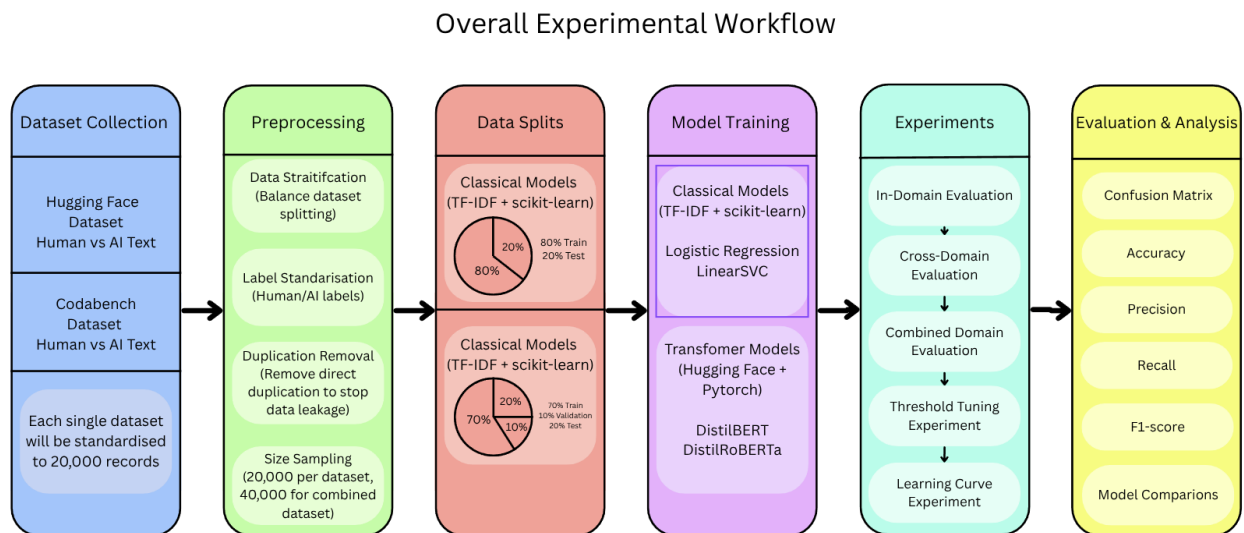


Figure 1: Overview of the experimental pipeline used in this project.

As shown in Figure 1, the overall implementation strategy for this research consisted of loading both the Codabench and Hugging Face datasets, preprocessing them to improve data quality, splitting the datasets into training and test sets, and training and evaluating both classical and transformer-based models. The Hugging-Face and Codabench datasets were specifically chosen for this project, due to their appropriate size, licensing and near-equal label distribution between Human and AI samples. Both of the datasets had been standardised to an equal size of 20,000 records to ensure that fair and consistent evaluations of model performance, and generalisability could take place. These models were evaluated across a variety of different conditions, such as in-domain, cross-domain and combined domain settings. In addition to this, the combined dataset was created and standardised to 40,000 records to evaluate the scalability and decision-making behaviour of models through threshold tuning and learning curve experiments. The results of each model were evaluated using standard classification metrics such as accuracy, precision, recall and F1-scores.

The preprocessing stage of this pipeline was conducted with slight variations tailored to each model type. For example, the classical models utilised a TF-IDF vectoriser to convert text samples into statistical features that can be processed by the models. Additionally, the datasets were divided into an 80% train and 20% test split. In comparison, the transformer models utilised a Hugging Face tokeniser to convert text into token IDs, which enables the models to capture contextual relationships between words in a sentence. Additionally, the transformer models utilised a 72% train, 8% validation and 20% test split across the dataset; each train-test partition was chosen to

facilitate reliable model training and performance evaluation. Despite these differences, all datasets were processed using the same cleaning methods, including label standardisation, sampling 20,000 records for each singular dataset experiment, and removing direct duplicates to prevent data leakage from occurring.

The classical models were implemented using the scikit-learn library. Logistic Regression and LinearSVC models were trained on the vectorised training set to generate predictions across the test sets. Logistic Regression and LinearSVC were selected as the final two classification models, as they can provide excellent baseline benchmarks due to their strong compatibility with TF-IDF vectorisation. Logistic Regression models can work well with linear relationships in datasets and have flexible probabilistic outputs that enable stronger analysis of the model's confidence. Alongside this, LinearSVC was chosen for its ability to handle complex decision boundaries; together, these two models offer simplicity and flexibility for conducting baseline experiments.

In contrast, fine-tuned transformer models are loaded using the Hugging Face library. Fine-tuned transformer models, such as DistilBERT, were chosen for this project because they are lightweight yet highly effective variants of BERT. In addition, DistilRoBERTa was chosen for its stronger general performance compared to DistilBERT, attributed to its improved training strategies. Both of these transformers were chosen over their heavier base models, as they provide faster training times whilst retaining strong performance. To improve reproducibility, once the datasets were tokenised, training was carried out using the Hugging Face trainer with predefined configurations, such as a fixed random seed, two training epochs to limit overfitting, fixed train/validation/test batch sizes, and a standardised evaluation strategy.

Each of these models underwent a range of experiments, including in-domain, cross-domain, and combined-domain generalisation testing. In-domain testing was chosen as the standard evaluation process because it assesses a model's ability to learn patterns from a dataset and can be used to measure performance on familiar data. In contrast, cross-domain testing has been utilised to evaluate a model's ability to generalise across new datasets. Furthermore, the combined dataset evaluation assesses how each model performs when trained on a mixed dataset, simulating real-world conditions. These experiments demonstrated whether a model has truly learned to distinguish between AI-generated and human-written text, or whether it has merely picked up on surface-level artefacts in the dataset. The transformers were further tested on extended combined-dataset experiments, such as threshold tuning and learning curves. These extended evaluations provided a comprehensive understanding of how transformers behave under varying thresholds and dataset sizes. These experiments were conducted under consistent settings and evaluated using standardised evaluation metrics, such as confusion matrices, precision, recall, accuracy, and F1-scores. These metrics enable a fair and strong comparative analysis across models.

Reliability, Validity, and Limitations of Methodology

The reliability of this study is maintained through the use of consistent experimental conditions across all model evaluations, including data stratification and consistent train/test splits. Both of the datasets used in this project have been standardised to 20,000 records, with an additional combined dataset consisting of 40,000 records used for extended experiments. Furthermore, the same evaluation metrics have been selected for all models, including accuracy, precision, recall, F1-score, and confusion matrices. Finally, the modular structure of both the classical and transformer pipelines supports reliability as it ensures all experiments can be tested again through the same conditions.

Furthermore, the validity of this study is displayed through the multiple sets of evaluative experiments such as in-domain, cross-domain, combined-domain, threshold tuning and learning curve tests. This project has not only tested models with familiar data, but also examined their ability to generalise across unseen data of different domains, therefore increasing the validity of the findings.

Finally, a key limitation that affects the overall scope of the project is the use of only two datasets alongside the exclusion of the original base transformers, such as BERT and RoBERTa. This limitation reduces the extent to which the study can evaluate more models and their ability to generalise across different datasets. While the addition of extra datasets and stronger transformer models could benefit the scope of the research, it would also result in increased computational costs and increased project management complexity due to the large number of experiments that already exist.

Results

For reader clarity, the names of the datasets have been abbreviated as follows: HF refers to the Hugging Face dataset, CB refers to the Codabench dataset, Combo refers to the Combined dataset. The notation $X \rightarrow Y$ refers to the training and testing dataset used for each experiment. For example, $HF \rightarrow HF$ means trained and tested on Hugging Face, while $CB \rightarrow HF$ means trained on Codabench and tested on Hugging Face.

Table 1: In-Domain Dataset Experiment Results

IN-DOMAIN DATASET EXPERIMENT					
Train $\rightarrow$ Test	Model	Accuracy	Precision	Recall	F1
$HF \rightarrow HF$	Logistic Regression	0.930	0.932	0.930	0.930
	LinearSVC	0.936	0.937	0.936	0.936
	DistilBERT	0.998	0.998	0.998	0.998
	DistilRoBERTa	0.997	0.997	0.997	0.997
$CB \rightarrow CB$	Logistic Regression	0.978	0.978	0.978	0.978
	LinearSVC	0.986	0.986	0.986	0.986
	DistilBERT	0.995	0.994	0.997	0.995
	DistilRoBERTa	0.998	0.999	0.999	0.999

Table 1 presents the findings from the in-domain experiment. In both datasets, fine-tuned transformer models performed better than classical models. For the $HF \rightarrow HF$ experiment, DistilBERT achieved the highest F1-score of 0.998, whilst the Logistic Regression performed the worst with an F1-score of 0.930. Furthermore, for the $CB \rightarrow CB$ experiment, DistilRoBERTa performed best with a near-perfect 0.999 F1-score, while the Logistic Regression model performed worst with a 0.978 F1-score. Across the classical models, LinearSVC consistently outperformed Logistic Regression, achieving F1-scores of 0.936 for $HF \rightarrow HF$ and 0.986 for $CB \rightarrow CB$. DistilRoBERTa also performed strongly on $HF \rightarrow HF$ with an F1-score of 0.997, whilst DistilBERT achieved 0.995 on $CB \rightarrow CB$.

Table 2: Cross-Domain Dataset Experiment Results

CROSS-DOMAIN DATASET EXPERIMENT					
Train → Test	Model	Accuracy	Precision	Recall	F1
HF→CB	Logistic Regression	0.599	0.704	0.599	0.539
	LinearSVC	0.646	0.748	0.646	0.605
	DistilBERT	0.499	0.500	0.994	0.665
	DistilRoBERTa	0.500	0.500	1	0.667
CB→HF	Logistic Regression	0.611	0.680	0.611	0.566
	LinearSVC	0.631	0.672	0.631	0.605
	DistilBERT	0.501	0.496	0.408	0.447
	DistilRoBERTa	0.742	0.923	0.522	0.667

Figure 2: Confusion Matrix for DistilRoBERTa in the HF→CB Setting

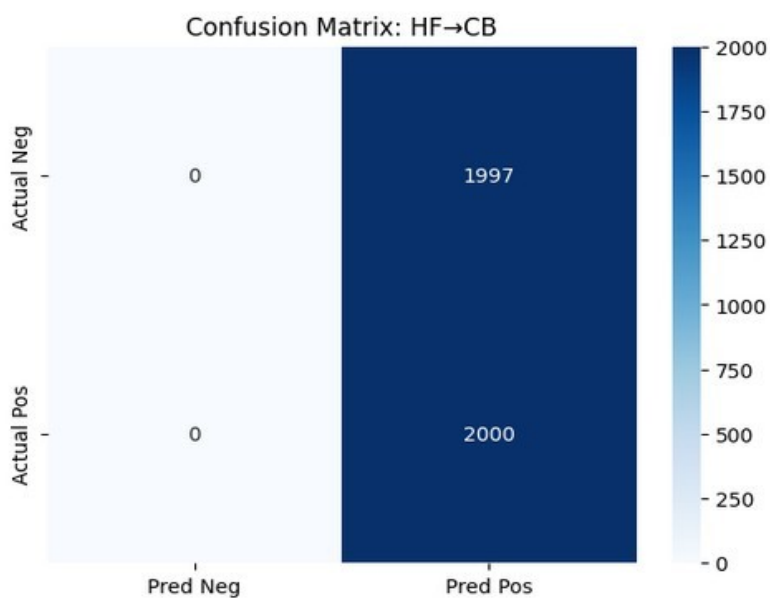


Table 2 presents the findings from the cross-domain dataset experiment. Overall, the models performed significantly worse in the cross-domain setting than in the in-domain setting. In HF→CB, DistilBERT achieved an F1-score of 0.665, whilst DistilRoBERTa recorded the highest F1-score of 0.667; this should be interpreted with caution, as the confusion matrix in Figure 2 suggests the model exhibited a complete one-class prediction bias. The worst performing model in HF→CB was Logistic Regression, with an F1-score of 0.539. In contrast, for CB→HF, DistilRoBERTa performed the best with an F1-score of 0.667, while DistilBERT performed the

worst with 0.447. Across the classical models, LinearSVC consistently performed better than Logistic Regression, for HF→CB (0.605 vs 0.539) and CB→HF (0.605 vs 0.566). Overall, these findings suggest that the models were vulnerable to domain shift and struggled to generalise across unseen data.

Table 3: Combined-Domain Dataset Experiment Results

COMBINED-DOMAIN DATASET EXPERIMENT					
Train → Test	Model	Accuracy	Precision	Recall	F1
Combo→Combo	Logistic Regression	0.946	0.947	0.946	0.946
	LinearSVC	0.955	0.955	0.955	0.955
	DistilBERT	0.993	0.990	0.996	0.993
	DistilRoBERTa	0.998	0.997	0.998	0.998
Combo→HF	Logistic Regression	0.947	0.949	0.947	0.947
	LinearSVC	0.978	0.978	0.978	0.978
	DistilBERT	0.997	0.997	0.997	0.997
	DistilRoBERTa	0.997	0.997	0.997	0.997
Combo→CB	Logistic Regression	0.982	0.982	0.982	0.982
	LinearSVC	0.996	0.996	0.996	0.996
	DistilBERT	0.989	0.983	0.996	0.990
	DistilRoBERTa	0.998	0.998	0.999	0.999

Table 3 presents the results of the combined dataset experiments. Overall, both fine-tuned transformers performed better than the classical models. DistilRoBERTa achieved the best performance with an F1-score of 0.998, followed by DistilBERT with 0.993. Among the classical models, Logistic Regression performed the worst across this experiment, with an F1-score of 0.946; LinearSVC followed closely with a slightly better F1-score of 0.955. A similar trend was observed for the Combo→HF settings. DistilRoBERTa and DistilBERT both remained the best performing models with F1-scores of 0.997, whilst the classical models performed the worst with LinearSVC and Logistic Regression performing with F1-scores of (0.978 vs 0.947). This trend slightly shifts for the Combo→CB setting, as DistilRoBERTa achieved a near-perfect F1-score of 0.999; however, this time LinearSVC follows with an F1-score of 0.996. This resulted in DistilBERT performing the second worst, with an F1-score of 0.990, whilst Logistic Regression performed the worst with 0.982. Overall, these results suggest that training in a combined-dataset setting improved every model's generalisability, as reflected by the high F1-scores. These findings show that DistilRoBERTa is presented as the strongest and most consistent model, while Logistic Regression is shown as the weakest overall.

Table 4: Combined Dataset Threshold Tuning Results

COMBINED-DOMAIN – THRESHOLD TUNING EXPERIMENT		DistilBERT				DistilRoBERTa			
Train → Test	Thresholds	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Combo → Combo	0.1	0.992	0.988	0.997	0.992	0.998	0.997	0.998	0.998
	0.3	0.993	0.989	0.996	0.993	0.998	0.997	0.998	0.998
	0.5	0.993	0.990	0.996	0.993	0.998	0.997	0.998	0.998
	0.7	0.994	0.991	0.996	0.994	0.998	0.998	0.998	0.998
	0.9	0.995	0.994	0.996	0.995	0.998	0.999	0.997	0.998

Table 4 presents results from the combined threshold-tuning experiment. Overall, both models' performance remained consistently high, with DistilBERT's F1-score ranging from 0.992 at a 0.1 threshold to 0.995 at a 0.9 threshold. DistilRoBERTa consistently achieved the same F1-score of 0.998 across all thresholds. A subtle yet clear trend is observed across the evaluation metrics as the threshold increases. For example, DistilBERT, when the threshold is set to 0.1, accuracy is 0.992, precision is 0.988, and recall is 0.997; however, when the threshold increases to 0.9, accuracy increases to 0.995, precision increases to 0.994, whilst recall decreases to 0.996. DistilRoBERTa displays a similar trend, precision increases from 0.997 to 0.999, whilst recall decreases from 0.998 to 0.997. Although the changes observed across metric values are small, these changes suggest that the models are very confident and therefore are largely unaffected by threshold variations.

Table 5: Combined-Domain Dataset Learning Curve Experiment Results

COMBINED-DOMAIN – LEARNING CURVE EXPERIMENT		DistilBERT				DistilRoBERTa			
Train → Test	Dataset Size	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Combo → Combo	10%	0.978	0.968	0.987	0.978	0.994	0.991	0.998	0.994
	25%	0.988	0.983	0.993	0.988	0.996	0.994	0.998	0.996
	50%	0.993	0.990	0.995	0.993	0.998	0.996	0.999	0.998
	100%	0.994	0.991	0.997	0.994	0.997	0.995	0.999	0.997

Figure 3: Combined-Domain Learning Curve Experiment Graph

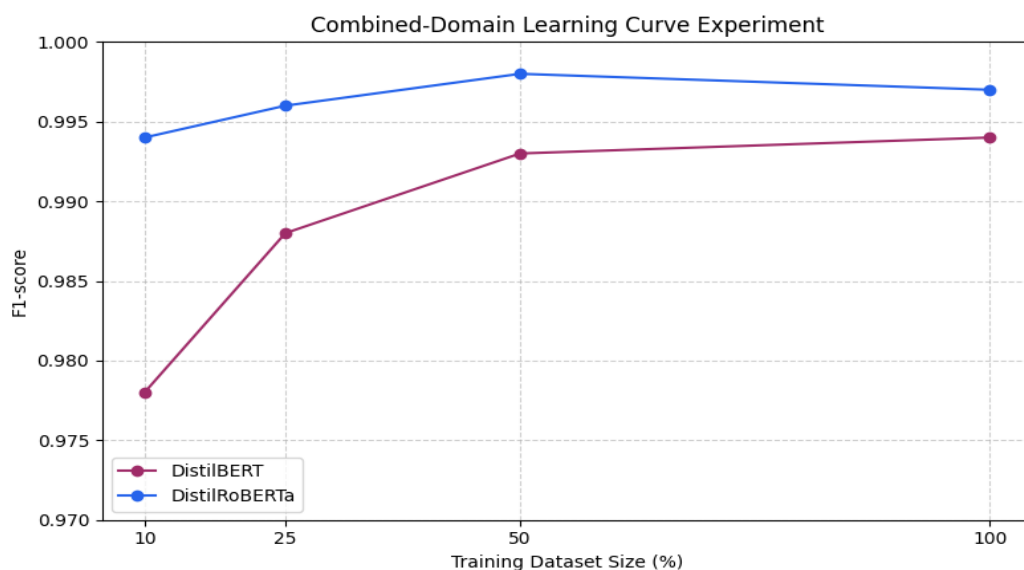


Table 5 reports the results from the combined-dataset learning curve experiment, which evaluates both transformers' performance across training data sizes ranging from 10% to 100%. Overall, the results suggest that increasing the training set size improves performance for both transformers, although these improvements level off once the training set exceeds 50%, as seen in Figure 3. For DistilBERT, all evaluation metrics steadily increase as the dataset size grows. For example, with a 10% size, DistilBERT achieved an accuracy of 0.978, precision of 0.968, recall of 0.987, and an F1-score of 0.978. With a dataset size of 50%, all evaluation metrics improve as accuracy increases to 0.993, precision to 0.990, recall to 0.995 and F1-score to 0.993. A similar trend was observed for DistilRoBERTa, which achieved metric values of 0.994 for accuracy, 0.991 for precision, 0.998 for recall, and 0.994 for F1-score at 10% of the dataset. When the training set size increased to 50%, the evaluation metric values also increased to 0.998 for accuracy, 0.996 for precision, 0.999 for recall and 0.998 for F1-score.

Evaluation

Evaluation of Results

Under in-domain settings, both classical and transformer-based models achieved relatively high or near-perfect F1-scores (classical: 0.930-0.986 vs transformers: 0.997-0.999). These results indicate that the models have reached a performance ceiling due to the low difficulty of the dataset, potentially capturing dataset-specific artefacts rather than truly learning the semantic differences between AI and human text. This observation is further supported by the marginal difference in performance between the DistilBERT and DistilRoBERTa models, indicating that introducing a model with increased complexity will likely yield minimal improvements across the in-domain experiments. Furthermore, the strong performance of both classical models demonstrates that relatively simple classical feature extraction processes are sufficient to achieve effective classification in this experiment, further reinforcing the observation that this classification task is relatively simple for the models. As a result, in-domain experiments alone are an unreliable measure of a model's true capability, as strong performance on familiar data does not necessarily reflect reliable generalisation across unseen domains.

Building on this observation, the cross-domain results show a significant decline in performance compared to the in-domain experiments. For example, compared to the in-domain F1-scores, DistilBERT decreased by 0.333 for HF→CB and 0.548 for CB→HF; similarly, DistilRoBERTa decreased by 0.330 and 0.332. This steep decline further supports the observation that the transformer models are learning dataset-specific artefacts rather than robust semantic features. Moreover, the significant drop in transformer performance compared to classical models suggests that whilst transformers have greater semantic representational capacity, they may be more vulnerable to superficial dataset-specific artefacts.

The performance collapse of the cross-domain experiment can be attributed to differences in style, length, and purpose between the two datasets. The difference between the average length of the AI records across the two datasets is quite substantial: the HF dataset averages approximately 100 words per text, whilst the CB dataset averages approximately 250 words. This suggests that length has become an unreliable artefact that the models have learnt. As the AI text in CB is 2.5x longer than the AI texts in HF, the models can learn to classify text based on sentence length, which can work well within their in-domain settings; however, they can fail to generalise across unseen domains. Furthermore, each dataset differs in its stylistic tone and structure. HF primarily maintains a task-oriented tone in its AI texts, which are based on assistant-style prompt-response outputs, whereas its human texts consist of encyclopedic articles from Wikipedia. In contrast, dataset CB is more structurally consistent, as both AI-generated and human texts adhere to news and academic writing styles. This structural consistency may explain why DistilRoBERTa generalised more successfully from CB → HF, scoring 0.742 in accuracy and 0.667 in F1-score, whereas HF → CB led to collapsed classification behaviour.

Furthermore, these results suggest that evaluation metrics can sometimes create a misleading description of model performance when considered in isolation, as significantly high F1 or Recall scores can hide unstable classification behaviour. As discussed previously, in HF→CB, the DistilRoBERTa model achieved a recall score of 1.00; however, it only achieved 0.5 in accuracy and precision. Similarly, DistilBERT showed a similar pattern, with a recall of 0.994 and an accuracy of 0.499. The F1-scores for both models remained around 0.665-0.667, even though they

showed significant class prediction bias. This is a clear case of metric inflation, where reliance on a single metric can inflate model performance, creating a false impression of a model's generalisability. This demonstrates why confusion matrices, such as Figure 2, are essential for visualising class prediction bias.

The diversity and coverage of a dataset affect model performance and generalisability more than a model's architecture complexity. This insight is supported by training all the models on an HF + CB combined dataset, which restored strong performance across the combined experiments, suggesting that exposure to a dataset containing a mixture of domains can reduce the dataset-specific artefacts that models pick up. This observation is further supported by the performance of the transformer models such as DistilBERT, which produced F1-scores ranging from 0.447 to 0.665 in cross-domain settings. Within the combined dataset, performance increased to 0.993 on Combo→Combo, 0.997 on Combo→HF, and 0.990 on Combo→CB. Similarly, the classical models display the same pattern, with the LinearSVC model, achieving F1-scores of 0.955, 0.978, and 0.996. These results indicate that broader domain coverage across datasets improves model generalisability more effectively than increased model complexity.

Increasing a model's training data improves its overall performance; the rate of improvement decreases over time as the training set approaches its full size. This relationship is observed with the transformer learning curve experiment. DistilBERT initially attains an F1-score of 0.978 at 10% training size; this increases to 0.988 at 25%, 0.993 at 50%, and finally 0.994 at 100%. This clearly shows that most improvements occur at an early stage between 10% and 50%. The minimal performance gains recorded between 50% to 100% training size indicate that the model's performance at this stage has plateaued, with additional training providing minimal further gains. DistilRoBERTa follows a similar pattern, starting with an F1-score of 0.994 at 10%, 0.998 at 50%, and 0.997 at 100%, suggesting that additional data provides limited value, as it maintains near-perfect performance across a wide range of data sizes. DistilRoBERTa's stronger low-data performance compared to DistilBERT could reflect its higher-quality pre-trained representations and objectives.

In addition, the threshold tuning experiment revealed that calibrating the threshold had subtle impacts on both transformer models. For example, the DistilBERT model achieved an F1-score of 0.992, a recall of 0.997, and a precision of 0.988 at a 0.1 threshold. Subsequently, as the threshold value increased to 0.9, the F1-score increased to 0.995, the recall to 0.996, and the precision to 0.994. This pattern is also seen in DistilRoBERTa's results: at a 0.1 threshold, recall and precision are 0.997 and 0.998, respectively; at a 0.9 threshold, precision increases to 0.999 while recall decreases to 0.997. These minimal changes suggest that as the threshold increases, the model becomes slightly stricter, resulting in reduced false positives while slightly increasing false negatives. Overall, these results indicate that the transformer models are highly confident in their predictions.

Overall, transformers achieve higher peak performance; however, classical models show slightly greater stability across domains. This is reflected in the near-perfect F1-scores achieved by transformers in in-domain settings (0.995-0.999), compared with classical models' F1-scores of 0.930-0.986, demonstrating the performance advantage that contextual representation learning provides. However, in cross-domain environments, transformers experience significant drops or failures compared to classical models. For example, both transformers displayed severe class

prediction bias in the HF→CB settings: DistilBERT achieved 0.499 accuracy, 0.500 precision, 0.994 recall, and 0.665 F1-score; similarly, DistilRoBERTa achieved 0.500 accuracy, 0.500 precision, 1.000 recall, and 0.667 F1-score. These results show a single-class prediction collapse rather than meaningful model generalisation. In contrast, although the classical models also perform weakly, they still appear to perform meaningful text classification, rather than collapsing into one-class prediction. In summary, whilst transformers can offer strong performance in a controlled environment, they are more vulnerable to picking up surface-level dataset patterns, suggesting that classical models can be slightly more robust under domain-shift settings.

Based solely on these experiment results, current detection models may not be reliable enough for real-world implementation, as their performance can be heavily affected by dataset similarity and domain shifts. The cross-domain results suggest a serious risk of genuine work being incorrectly flagged as AI. Incorrect classifications within academic institutions can undermine integrity and lead to unfair consequences. Furthermore, the combined dataset experiments suggest that a real-world implementation would require large and diverse training sets to achieve reliable performance.

Evaluation of Objectives

To understand how effectively each research objective has been addressed, each research question previously established in the methodologies section will be answered with reference to the results obtained from each experiment.

The first research question was met, as transformer models outperformed the classical models for the in-domain and combined-domain settings, as they were able to achieve near-perfect F1-scores displaying their advantage in performance due to their contextual representation capabilities.

However, the results show transformers to be generally less stable than classical models in cross-domain settings, as they are vulnerable to dataset-specific artefacts.

The second research question was met, as all the models generalised very poorly across the cross-domain settings. The transformers in the HF→CB settings displayed misleading performance, as their overall F1-score was inflated by severe one-class prediction bias. In contrast, while classical models also performed poorly, they displayed stable prediction behaviour, suggesting that they were still able to maintain balanced classification predictions, even under domain shift.

The third research question was met, as the results indicated that increasing the dataset training size improved each transformer model's performance. Once these training sizes exceeded 50%, the rate of improvement levelled off.

The fourth research question was met, as the results suggested that threshold tuning had a minimal yet identifiable effect on each transformer model's performance. For example, at lower thresholds, precision would decrease while recall increased. In contrast, at higher thresholds, precision would increase, while recall decreased. However, as these changes were marginal, they suggest that the models were already highly confident in their predictions.

Lastly, error patterns suggest that the strongest-performing model, DistilRoBERTa, achieved a near-perfect score across in-domain settings; however, it collapses under domain shift and learns surface-level patterns and shortcuts. In contrast, the worst-performing model, Logistic Regression, achieved a weaker overall performance but was slightly more stable across distribution shifts, as it was less vulnerable to picking up dataset artefacts.

Research Validity

The validity of this research project is supported by multiple test conditions established to achieve the aim of understanding and comparing the performance of AI text detection models. These experimental conditions include in-domain, cross-domain, and combined dataset testing. Furthermore, additional transformer-specific model tests, such as threshold tuning and learning curve experiments, have been conducted to evaluate model behaviour across varying decision boundaries and training data sizes. However, validity is also limited by inherent differences between the HF and CB datasets. The models may be learning surface-level patterns that affect their ability to generalise. For example, the CB dataset's AI texts are, on average, 2.5x longer than those in the HF dataset. There are also stylistic differences in the AI samples: HF contains assistant-style prompt-response outputs, whilst CB contains AI-generated academic/news articles. The effects of these differences are clearly seen in the strong in-domain performance of DistilRoBERTa (0.997-0.999), which severely declines in cross-domain settings, leading the model to exhibit a one-class prediction bias. However, this limitation in validity was partially addressed by training each model on combined datasets to improve generalisation.

Research Reliability

Furthermore, the reliability of each experiment is supported, as each has been carried out with consistent settings to ensure a fair comparison across models. Examples of these configurations include: all datasets have been preprocessed using the same pipeline, such as stratified splitting, class balancing, label standardisation, removal of duplicate data from train/test sets, and standardising dataset sizes to 20,000 and 40,000 records. Furthermore, reliability is supported by the consistently strong performance output by each model across the in-domain and combined dataset settings. Moreover, the threshold-tuning and learning-curve experiments have shown smooth, expected behaviour from the transformer model. However, the reliability of this study is limited in cross-domain settings, where model performance has been unstable, leading to a one-class prediction bias for both transformer models in HF→CB settings. Overall, the reliability of model behaviour can be supported in controlled settings; however, it cannot be assumed to extend to unseen datasets.

Research & Methodological Limitations

This project is limited by the scope of the two datasets utilised across the experiments; as a result, the ability to measure model generalisability across broader real-world domains has been limited. Furthermore, datasets CB and HF differ significantly in structure and style. Whilst dataset HF consists of shorter, encyclopedic human samples and assistant-style responses for the AI samples, dataset CB contains long-form academic and news-style samples; as a result, this may introduce dataset bias. A further limitation is the restricted selection of transformer models. Although DistilBERT and DistilRoBERTa are advantageous due to their reduced computational cost and training efficiency, including larger full-size base models such as BERT and RoBERTa would have provided a stronger foundation for a comprehensive comparison of model capabilities between the lighter models and the full-size base models. Additionally, as this project primarily focuses on binary classification between AI and human text, it is limited in its ability to reflect real-world complexity, as real-world text origin is not exclusively split into AI or human classes. For example, text can often be paraphrased and edited by humans or generated by multiple different LLMs.

A primary limitation across the experimental design of this research was limited control over dataset-specific artefacts; as a result, these models may have picked up on cues such as differences in structure, style, and length across the two datasets. Furthermore, whilst direct duplicates across datasets have been removed, no mechanisms were employed to detect and remove highly similar texts, which could affect model performance. Lastly, the experimental pipeline is further limited by

the lack of additional hyperparameter exploration. While all the models were trained with fixed training configurations to maintain consistency, additional hyperparameters, such as class weightings or learning rate schedules, could have improved the models' ability to generalise across the experiments where they struggled.

Reflection

Throughout this project, it was initially expected that the classical models would perform poorly. In contrast, fine-tuned transformers would be highly effective under all conditions, regardless of factors such as dataset domain and variation. However, in reality, the project findings showed that transformers can be highly susceptible to domain shift. The domain shift was a primary challenge: because the datasets were so different, the transformer models picked up dataset-specific artefacts, such as length, style, and structure, which can significantly affect model performance. As a result, in HF→CB settings, both transformer models exhibited one-class bias, leading to confusing and inflated F1-scores. These challenges highlighted the importance of including diverse, well-balanced datasets for model training and the need for appropriate hyperparameter configuration to reduce the likelihood of models picking up dataset-specific artefacts and improve generalisation. Furthermore, this research highlighted the importance of considering multiple evaluation metrics, rather than relying on a single score to capture a model's behaviour. This project enabled the development of various skills, including building and applying NLP pipelines, configuring text classification models, and developing systematic data analysis skills.

Future Work

Future classification projects will use datasets with greater diversity, spanning a wider range of domains. This would enhance the model's ability to generalise across different domains. Furthermore, more mechanisms will be implemented to control dataset artefacts, such as differences in text length, stylistic patterns, and near-duplicate samples. These mechanisms will ensure that the models are less susceptible to picking up surface-level patterns that harm model generalisability. Moreover, future text classification projects will also aim to incorporate stronger domain-adaptation methods such as Task-Adaptive PreTraining. Future classification work will also aim to involve an increased variety of hyperparameter tuning, such as learning rate schedules, model weight decays and class weightings. This may improve a model's ability to perform and generalise across unseen domains. In addition, full-size base models such as BERT and RoBERTa should also be implemented to enable more in-depth comparative evaluations of teacher and student model behaviour under the same conditions. Finally, the model evaluation process should be improved by including additional methods such as precision-recall curves and testing models on real-world literature such as student essays or journal papers.

Conclusion

In conclusion, the primary objective of this project was to compare and evaluate the performance of classical and fine-tuned transformer models in AI text classification. The models were evaluated using two different datasets and multiple experimental settings. Overall, the results of these experiments indicated that while fine-tuned transformer models such as DistilBERT and DistilRoBERTa can achieve near-perfect performance under in-domain settings, their performance can significantly decrease when they are evaluated across unseen cross-domain settings. This suggests that the transformer models were affected by domain shift. This may be attributed to their potential vulnerability to generalisation shortcuts, such as differences in sentence length and style between human-written and AI-generated samples. Whilst these dataset-specific artefacts can be effective within the in-domain setting, they can struggle to transfer across cross-domain conditions. This behaviour was observed in the HF→CB experiment, where each model displayed one-class

prediction bias. In contrast, classical models such as LinearSVC and Logistic Regression showed overall lower performance across most experiments, however, when they were tested under cross-domain settings such as HF→CB, their behaviour remained comparatively more stable than the fine-tuned transformers. This may suggest that they were less vulnerable to the dataset-specific artefacts and that they attempted to genuinely classify the text samples. Notably, when domain adaptation was introduced through the combined dataset, the transformer models regained their strong near-perfect performance. This suggests that exposure to a more diverse training set can enable the models to increase their generalisability. This raises a key insight as it appears that the characteristics and diversity of a dataset can have a greater impact on model performance than a model's complexity.

Unexpected results, such as the one-class prediction bias exhibited by DistilRoBERTa in the HF→CB settings, highlight a significant limitation of AI text detection in real-world applications, especially in high-stakes environments such as academic institutions, where false-positive classifications can incorrectly label genuine work as AI-generated, leading to unfair consequences. Ultimately, current AI text classification models can perform strongly under controlled experiments, however, in real-world environments their reliability can be limited as texts are often noisy and subject to evasion techniques such as paraphrasing. To address these challenges, AI detection models will need to be trained on more diverse real-world datasets, integrate stronger mechanisms that reduce over-reliance on dataset-specific artefacts and implement robust evaluation methods that offer a deeper understanding of model behaviour.

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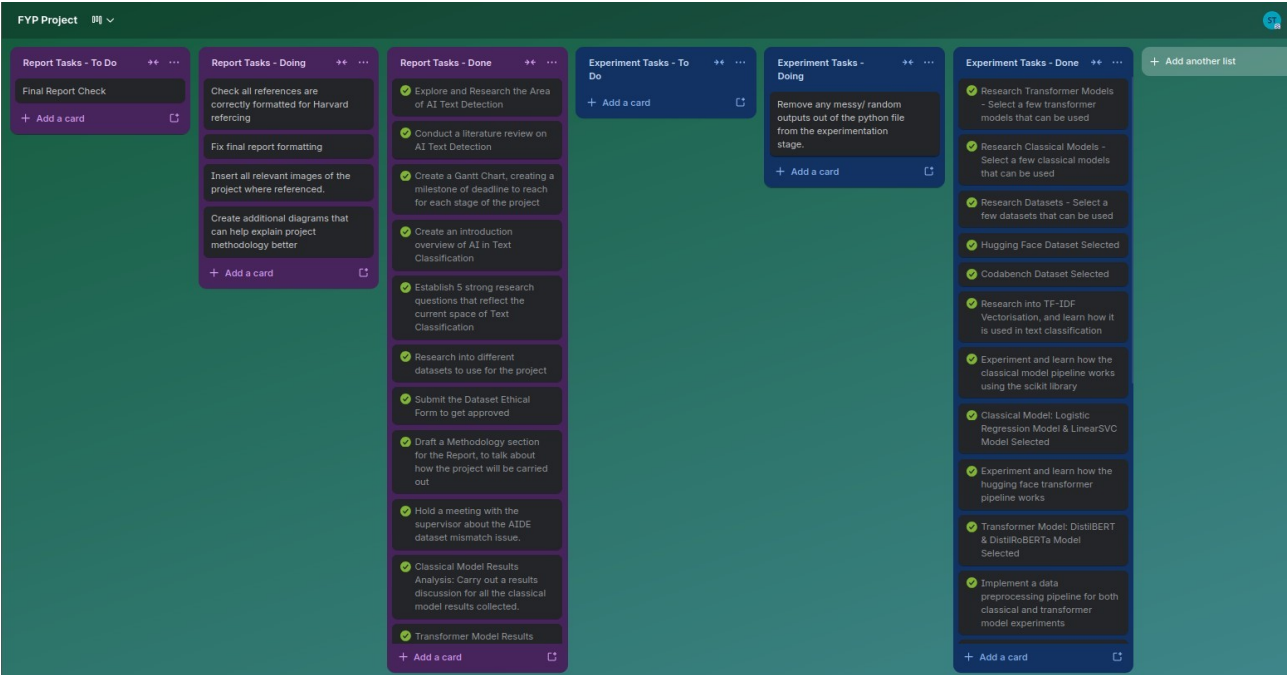
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Appendices

Appendix A

Figure A1: Trello Board



Appendix B

Figure B1: Old Gantt Chart

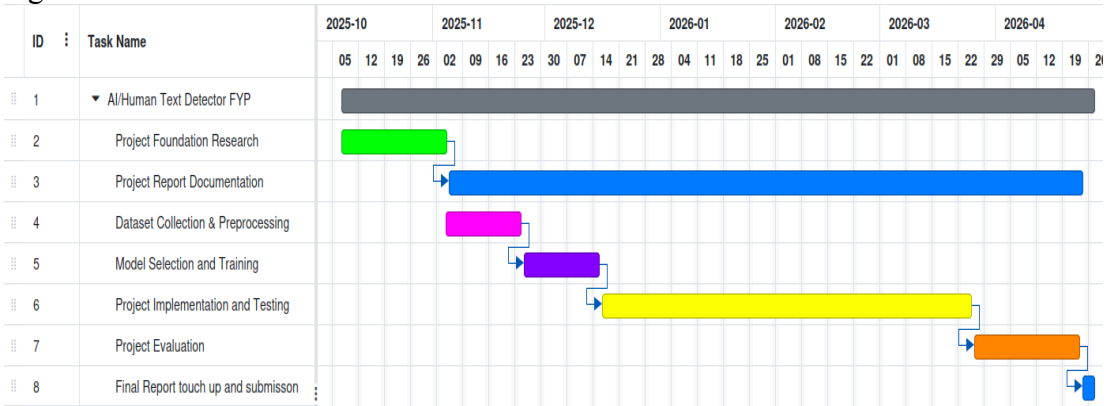


Figure B2: New Gantt Chart

